

Supplementary Information

Methods

1 Observational reanalysis datasets

2 Four independent global ocean reanalysis products are analysed. They span distinct ocean-model fami-
3 lies, horizontal resolutions, assimilation schemes, and surface-forcing choices, and together represent the
4 currently available space of observationally-constrained three-dimensional ocean state estimates at 34.5°S.

5 **ORAS5** [Zuo et al., 2019]. Ocean Reanalysis System 5 from ECMWF. Native grid: ORCA025 tripolar
6 (nominal $\sim 1/4^\circ$, ~ 25 km at the equator and ~ 14 km at 34.5°S after the $\cos \phi$ contraction), 75 vertical
7 levels with 1-m surface resolution. NEMOVAR 3D-Var FGAT data-assimilation scheme ingesting sub-
8 surface temperature/salinity profiles, sea-ice concentration, and along-track altimetric sea-level anomalies.
9 Atmospheric forcing: ERA-40 (1958–1978) transitioning to ERA-Interim/ERA5 (1979–present). Record
10 used: 816 monthly means, 1958–2025.

11 **GLORYS12V1** [Lellouche et al., 2021]. Copernicus Marine Service/Mercator Ocean global ocean
12 reanalysis at $1/12^\circ$ (~ 8 km, eddy-resolving), 50 vertical levels. A reduced-order Kalman filter (SAM2)
13 jointly assimilates along-track altimetry, satellite sea-surface temperature, sea-ice concentration, and in-
14 situ CTD and Argo temperature/salinity profiles, supplemented by 3D-Var bias correction. Atmospheric
15 forcing: ERA5. Record used: 396 monthly means, 1993–2025 (satellite-altimetry era).

16 **SODA 3.15.2** [Carton et al., 2018]. Simple Ocean Data Assimilation 3 from the University of Maryland.
17 Native grid: MOM5 $\sim 1/2^\circ$ (regrided to a uniform 0.5° product on the UMD mirror), 50 vertical levels.
18 Assimilation scheme: ensemble optimal interpolation with a 5-day analysis cycle; the downloadable output
19 is a sequence of 5-day snapshots. Atmospheric forcing: MERRA-2. We sample four 5-day snapshots per
20 year (mid-month of January, April, July, October) to approximate the annual mean, consistent with
21 previous applications of the same product to overturning-transport diagnostics. Record used: 43 annual
22 means, 1980–2022.

23 **ECCO-V4r4** [Forget et al., 2015]. NASA/JPL Estimating the Circulation and Climate of the Ocean,
24 version 4 release 4. Native grid: LLC90 Lat-Lon-Cap (regrided to a uniform 0.5° monthly product), 50
25 vertical levels. ECCO is an adjoint-based least-squares state estimate (4D-variational) that iteratively
26 adjusts initial conditions, surface fluxes, and interior mixing to minimise the misfit between the model
27 trajectory and the full suite of altimetry, Argo, satellite SST/SSH, and mooring observations over its full
28 time span. Atmospheric forcing (post-iteration): an ECCO-adjusted ERA-family atmospheric analysis.
29 Record used: 26 annual means, 1992–2017.

30 **Direct-hydrography anchors.** Three published overturning freshwater-transport estimates at 34.5°S
31 from direct in-situ observations are used as independent anchor points in Fig. 1: Garzoli and Matano
32 [2011] (central year 2005, value -0.10 ± 0.10 Sv); Meinen et al. [2018] (central year 2015, value $-0.09 \pm$
33 0.05 Sv); and Kersalé et al. [2020] (central year 2015, value -0.17 ± 0.07 Sv). Together with Dong et al.
34 [2024]’s -0.15 ± 0.09 Sv from 49 XBT transects, they bracket the reanalysis ensemble from the low side.

35 CMIP6 ensemble

36 CMIP6 meridional-velocity (**vo**) and salinity (**so**) sections at 34.5°S are obtained for every available
37 model with coverage of both the **historical** experiment (required period 1950–1980) and the **ssp585**
38 experiment (required period 2080–2100). The native grid of each model is used (i.e. no regridding),
39 because regridding distorts the mechanistic decomposition. Fifteen models satisfy both criteria with

accessible section files: CESM2, CMCC-CM2-SR5, CNRM-CM6-1, CanESM5, FGOALS-g3, FIO-ESM-2-0, GFDL-CM4, GISS-E2-1-G, HadGEM3-GC31-LL, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, MPI-ESM1-2-LR, NESM3, UKESM1-0-LL. Three models are excluded: NorESM2-LM and NorESM2-MM because their ocean component is on isopycnal (density) coordinates and cannot be integrated in depth-space without an additional remap that would confound the decomposition; and ACCESS-CM2 because its accessible `spp585` run does not extend to 2100. All models use the `r1i1p1f1` realisation.

A 13-model subset with `piControl` access is used for the internal-variability null test: ACCESS-CM2, CESM2, CMCC-CM2-SR5, CNRM-CM6-1, CanESM5, GISS-E2-1-G, HadGEM3-GC31-LL, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, MPI-ESM1-2-LR, NESM3, UKESM1-0-LL.

Section extraction and grid metrics

At each time step of each product we extract the zonal-depth section at the gridded latitude closest to -34.5° . Where the ocean-model grid is staggered (e.g. velocity at U-points and salinity at T-points on the ORAS5 NEMO grid), the two fields are extracted on their native grids and aligned by truncation to the common Atlantic-longitude mask.

The Atlantic basin at 34.5°S is defined by the longitude range $[-70^\circ, 20^\circ]$ in the $[-180^\circ, 180^\circ]$ convention (bounds chosen inclusively to cover from the Argentine coast to the African coast while excluding residual Agulhas leakage; the small $< 1\%$ contribution from the strictly-classified Drake-Passage edge does not alter the results).

The zonal grid spacing at each Atlantic U-point is

$$\Delta x_i = |\Delta \lambda_i| R_\oplus \cos \phi,$$

with $\Delta \lambda_i$ the local longitudinal grid spacing (after longitude wrap-around correction), $R_\oplus = 111$ km per degree at the equator, and $\phi = -34.5^\circ$. Vertical cell thicknesses Δz_k are taken directly from the product's vertical coordinate (with the CESM-family POP grid converted from centimetres to metres).

F_{ovS} computation

We follow the definition of de Vries and Weber [2005], used also by Dijkstra [2007], Drijfhout et al. [2011], van Westen et al. [2024]:

$$F_{\text{ovS}} = -\frac{1}{S_0} \int_{z_{\text{bot}}}^{z_{\text{surf}}} V_{\text{int}}(z) [\bar{S}(z) - S_0] dz,$$

with

$$V_{\text{int}}(z) = \sum_{i \in \text{Atl}} v(x_i, z) \Delta x_i \quad (\text{m}^2/\text{s}),$$

and

$$\bar{S}(z) = \frac{\sum_{i \in \text{Atl}} S(x_i, z) \Delta x_i \cdot m(x_i, z)}{\sum_{i \in \text{Atl}} \Delta x_i \cdot m(x_i, z)},$$

where $m(x_i, z) \in \{0, 1\}$ is the wet-cell mask (unity over the ocean, zero over bathymetry). The reference salinity is fixed at $S_0 = 35$ PSU throughout. The integral is evaluated by a discrete sum over vertical levels using the native Δz_k as quadrature weights; the final result is reported in Sverdrups ($\text{Sv} = 10^6 \text{ m}^3/\text{s}$). This kernel is implemented once in `ardp/physics/fovs.py` and reused for every product (reanalyses, CMIP6 historical/spp585, piControl) to eliminate cross-product inconsistencies.

Barotropic-velocity subtraction

F_{ovS} is, by construction, the freshwater transport carried by the zonally-averaged *overturning* (baroclinic) circulation, i.e. the component with $\int V_{\text{int}}(z) dz = 0$ across the section. In a mass-conserving model the net volume transport through a zonal section is exactly zero, so this constraint is satisfied automatically. Data-assimilating Boussinesq reanalyses (ORAS5, GLORYS12V1, SODA 3.15.2, ECCO-V4r4) and Boussinesq coupled models in CMIP6, however, do *not* enforce volume conservation across an arbitrary zonal section, and their net transport $V_{\text{net}} = \int V_{\text{int}}(z) dz$ generally takes non-zero values that drift slowly between months and decades as data-assimilation increments or parameterised surface sources re-balance the basin. If this drift is left in, it couples to the offset $\bar{S}(z) - S_0$ and injects a spurious mSv-scale signal into F_{ovS} that has nothing to do with the overturning.

We therefore follow de Vries and Weber [2005], Mecking et al. [2017], Weijer et al. [2019] and explicitly subtract the section-mean (barotropic) velocity

$$\bar{v}_{\text{bt}} \equiv \frac{\int V_{\text{int}}(z) dz}{\int A_{xy}(z) dz}, \quad A_{xy}(z) \equiv \sum_{i \in \text{Atl}} \Delta x_i m(x_i, z),$$

before evaluating F_{ovS} , replacing $V_{\text{int}}(z)$ by its baroclinic residual $V_{\text{int}}^{\text{bc}}(z) \equiv V_{\text{int}}(z) - \bar{v}_{\text{bt}} A_{xy}(z)$. By construction, $\int V_{\text{int}}^{\text{bc}}(z) dz \equiv 0$, which eliminates the volume-drift artefact. The subtraction is applied independently to each monthly snapshot of each product (so the correction tracks the DA increments in time) and is enforced identically on all four reanalyses, on all CMIP6 models, and on the `piControl` null ensemble.

Across the two decomposition windows reported here the magnitude of V_{net} is ~ 0.9 – 1.4 Sv for ORAS5 and GLORYS12V1, ~ 1.1 – 1.3 Sv for ECCO-V4r4, and ~ 3.8 – 5.0 Sv for SODA 3.15.2 (see Table 1 in main text); the ~ 10 – 40 mSv spurious contribution that the correction removes is therefore of the same order as the $|\Delta F_{\text{ovS}}|$ signals that drive the mechanism classification, and its omission would have qualitatively flipped the classification in several cases. An independent unit test in `tests/test_fovs_decomposition.py` verifies the $\Delta F_{\text{ovS}} = 0$ identity for a pure uniform-velocity drift between two periods.

Mechanism decomposition

Define $\Delta F_{\text{ovS}} \equiv F_{\text{ovS}}(t_2) - F_{\text{ovS}}(t_1)$ for two non-overlapping period averages t_1 (“early”) and t_2 (“late”). Writing $v_2 \equiv v_1 + \Delta v$ and $\bar{S}_2 \equiv \bar{S}_1 + \Delta \bar{S}$ and substituting into the definition of F_{ovS} gives the exact identity

$$\Delta F_{\text{ovS}} = \Delta F_v + \Delta F_s + \Delta F_{\text{cross}},$$

where

$$\begin{aligned} \Delta F_v &= -\frac{1}{S_0} \int \Delta V_{\text{int}}(z) [\bar{S}_1(z) - S_0] dz, \\ \Delta F_s &= -\frac{1}{S_0} \int V_{\text{int},1}(z) \Delta \bar{S}(z) dz, \\ \Delta F_{\text{cross}} &= -\frac{1}{S_0} \int \Delta V_{\text{int}}(z) \Delta \bar{S}(z) dz. \end{aligned}$$

ΔF_v is the change that would have occurred if only the velocity field had changed (salinity frozen at its t_1 value); ΔF_s is the change that would have occurred if only the zonal-mean salinity had changed (velocity frozen at t_1); and ΔF_{cross} is the cross (“eddy”) term that is zero when Δv and $\Delta \bar{S}$ are exactly orthogonal in a depth-integrated sense and becomes large and signed when the two anomalies are correlated in z . The identity is exact up to floating-point round-off; unit tests verify residual magnitudes of $\mathcal{O}(10^{-15})$ Sv across randomised synthetic sections, velocity-only perturbations, salinity-only perturbations, and land-masked cases.

The *velocity share* and *salinity share* reported throughout are

$$f_v \equiv 100 \cdot \Delta F_v / \Delta F_{\text{ovS}}, \quad f_s \equiv 100 \cdot \Delta F_s / \Delta F_{\text{ovS}}, \quad f_c \equiv 100 \cdot \Delta F_{\text{cross}} / \Delta F_{\text{ovS}},$$

with the exact constraint $f_v + f_s + f_c = 100\%$. An individual share can exceed 100% or be negative whenever the three components do not all carry the same sign. This occurs naturally in the salt-advection-feedback regime when Δv and $\Delta \bar{S}$ are anti-correlated in the vertical (e.g. a slowed upper-limb circulation combined with a freshened subtropical layer); then ΔF_{cross} becomes large and of opposite sign to $\Delta F_v + \Delta F_s$, so that $|f_v|$ and $|f_s|$ each exceed 100%. Values $|f_v| > 100\%$ therefore indicate a physically meaningful cross-term, not a numerical artefact.

The decomposition becomes numerically ill-conditioned when $|\Delta F_{\text{ovS}}|$ is small relative to $|\Delta F_v|$, $|\Delta F_s|$, or $|\Delta F_{\text{cross}}|$. We pre-registered a threshold of $|\Delta F_{\text{ovS}}| \geq 10$ mSv below which the mechanism decomposition is declared *ill-defined*. ECCO-V4r4 is the only reanalysis flagged under this rule.

Period choice

The early and late periods were pre-registered before processing the fourth reanalysis (SODA 3.15.2): $t_1 = 1993$ – 2005 and $t_2 = 2013$ – 2025 . These are the longest non-overlapping windows fully within the common GLORYS12V1 record, avoiding the first few years of the satellite-altimetry era (pre-1993) while

preserving a $\gtrsim 10$ -year separation. For products with shorter records we use the nearest-in-length late window: SODA 3.15.2 uses $t_2 = 2013\text{--}2020$; ECCO-V4r4 uses $t_2 = 2013\text{--}2017$. For CMIP6 we pair historical 1950–1980 against **ssp585** 2080–2100 to maximise the forced signal-to-noise ratio.

The sensitivity of the classification to the period choice is reported in Supplementary Fig. S2 by repeating the computation across a grid of $(t_{1,\text{end}}, t_{2,\text{start}})$ pairs ($t_{1,\text{end}} \in \{2001, 2003, 2005, 2007, 2009\}$, $t_{2,\text{start}} \in \{2011, 2013, 2015, 2017, 2019\}$, always with $t_{1,\text{start}} = 1993$ and $t_{2,\text{end}}$ at each product’s data end). The resulting 25-point distribution of f_v values is reported per product.

Baseline regime classification

Beyond the mechanism decomposition, each product and CMIP6 model is classified by its *baseline* regime at the t_1 window: *bistable* if $\overline{F_{\text{ovs}}}(t_1) < 0$ (salt-advection feedback active, Rahmstorf [1996]) or *monostable* if $\overline{F_{\text{ovs}}}(t_1) > 0$. This matters because the salt-advection feedback is not physically active in the monostable regime, so a mechanism-classification result from a monostable CMIP6 model is not directly comparable to an observational reanalysis in which the Atlantic is bistable. In our ensemble 6/15 CMIP6 models are bistable at baseline (CNRM-CM6-1, CanESM5, IPSL-CM6A-LR, MIROC6, MPI-ESM1-2-HR, NESM3); 9/15 are monostable — the well-known “CMIP6 fresh bias” at the South Atlantic [Mecking et al., 2017, Weijer et al., 2019, Portmann et al., 2026].

Trend significance

Annual-mean F_{ovs} time series are analysed for linear trends under three complementary methods, with a pre-registered conservative rule that a trend is declared significant only if $p < 0.05$ under all three:

1. **Naive OLS.** Standard ordinary least-squares regression with residual-variance-based p -value, no autocorrelation correction. Reported as the upper-bound significance claim.
2. **Santer et al. [2000] N_{eff} .** The residual lag-1 autocorrelation r_1 is computed from the OLS residuals of the detrended series. The effective sample size is $N_{\text{eff}} = N(1 - r_1)/(1 + r_1)$; the trend’s standard error is multiplied by $\sqrt{N/(N_{\text{eff}} - 2)}$ and the adjusted p -value uses the Student- t distribution with $N_{\text{eff}} - 2$ degrees of freedom. We clip r_1 to $[-0.99, 0.99]$ for numerical stability.
3. **GLS Prais-Winsten AR(1).** The series is pre-whitened using a first-order autoregressive estimate of the residual autocorrelation (iterated Prais-Winsten), then the transformed series is fit by OLS; the resulting trend estimate and standard error explicitly account for AR(1) residual correlation.

Mechanism classification rule

A product is labelled *velocity-dominant* if $f_v > 60\%$, *salinity-dominant* if $f_s > 60\%$, *mixed* if both $f_v \leq 60\%$ and $f_s \leq 60\%$, and *ill-defined* (no classification) if $|\Delta F_{\text{ovs}}| < 10$ mSv. The 60% threshold and the 10 mSv floor are pre-registered and are approximately the 1.5σ and 2σ boundaries of the piControl null distribution (Supplementary Fig. S4). We also ran the analysis at 50%, 55%, and 65% classification thresholds and at 5 mSv and 20 mSv ill-defined cut-offs, with identical qualitative conclusions (Supplementary Fig. S2).

Outlier handling

For each annual F_{ovs} time series we flag annual values with $|F_{\text{ovs}}| > 0.3$ Sv as likely data artefacts, because all published estimates at 34.5°S lie within $[-0.2, +0.2]$ Sv and values beyond this threshold are $\gtrsim 3\sigma$ outside the observational band [Weijer et al., 2019]. Only one annual value across the four-product ensemble is flagged by this rule: SODA 3.15.2 in 1998, with $F_{\text{ovs}} = -1.77$ Sv. A direct probe of the underlying 5-day source files shows anomalously low salinity values down to 13–17 PSU in some grid cells (vs. the 34.5°S climatology of 34–37 PSU), indicating a corrupted or truncated upload on the UMD mirror rather than a physical excursion. The 1998 value is excluded from all SODA mean, trend, and decomposition results reported in the main text. Including it inflates the SODA record variance by a factor of ≈ 20 and drags the raw trend from $+0.6$ mSv yr^{-1} (n.s.) to $+1.4$ mSv yr^{-1} (also n.s.); the sign, significance, and mechanism attribution are all unchanged (see Supplementary Discussion).

piControl null test

For each of the 13 CMIP6 models with accessible piControl section data, we draw 200 pairs of random non-overlapping 30-year windows from the full piControl integration, compute the decomposition for each pair, and pool the results ($\leq 13 \times 200 = 2600$ decomposition samples in total, with some pairs rejected when the time series is too short). The resulting distribution of $|\Delta F_{\text{ovs}}|$ quantifies the internal-variability magnitude of the decomposition signal in the absence of any forcing change. A forced-experiment $|\Delta F_{\text{ovs}}|$ must exceed the 95th percentile of this null (22 mSv in our pooled ensemble, Supplementary Fig. S4) to be declared above internal variability. For the three diagnosable reanalyses, ORAS5 (31 mSv) and GLORYS12V1 (87 mSv) comfortably exceed the null 95th percentile, while SODA 3.15.2 (14 mSv) is above the null median but below the 95th percentile, consistent with the mixed classification we assign it.

Mechanism-conditional projection

In Fig. 4 of the main text we partition the 12 CMIP6 forced-weakening models by their mechanism class (v-dominant, s-dominant, mixed) and aggregate their ssp585 AMOC-at-26.5°N projections by class. For each model m we compute the percent AMOC weakening by 2100 as $100 \cdot (1 - A_m^{2081-2100} / A_m^{1950-1980})$, where A_m denotes the annual-mean overturning streamfunction at 26.5°N maximum below 500 m. The median and interquartile range per class are reported in the main text and visualised as boxplots. Class-ensemble mean AMOC trajectories are drawn only for years where every member of that class has data, avoiding misleading tails from 1–2 late-ending models.

Lead-lag cross-correlation analysis

To quantify the temporal relationship between the overturning freshwater transport F_{ovs} at 34.5°S and the AMOC strength at 26.5°N, we compute the cross-correlation function for each CMIP6 model individually. Annual time series of F_{ovs} and the maximum overturning streamfunction below 500 m at 26.5°N are detrended by removing the linear trend over the historical (1950-2014) and future (2015-2100) periods. The cross-correlation is then evaluated for lags from -20 to $+20$ years, where positive lag indicates that F_{ovs} leads AMOC. The peak correlation lag is recorded for each model. The significance of the correlation is assessed using a two-sided Student's t -test with effective degrees of freedom adjusted for autocorrelation [Santer et al., 2000].

Emergent constraint on future F_{ovs}

We follow the standard emergent constraint approach [Portmann et al., 2026, Dong et al., 2024]. For each CMIP6 model, we compute the historical mean F_{ovs} over the period 2000-2024 (the most recent decade common to all reanalyses) and the future mean F_{ovs} over 2081-2100 under SSP5-8.5. A least-squares linear regression is fitted to the model ensemble, with historical F_{ovs} as the predictor and future F_{ovs} as the predictand. The observational estimate of historical F_{ovs} (mean of the four reanalyses, $\bar{F}_{\text{ovs}} = -0.078 \pm 0.024$ Sv) is then projected onto the regression line to obtain a constrained future value. Confidence intervals (95%) are derived using the standard error of the prediction, accounting for the spread in the model ensemble and the uncertainty in the observational mean.

Software and reproducibility

All analyses are implemented in Python 3.13 using NumPy, SciPy, xarray, dask, pandas, matplotlib, and the cross-label-placement package adjustText. CMIP6 catalogue access uses intake-esm and pangeo-cmip6; NASA Earthdata access uses earthaccess. The package ardp contains the shared kernels (fovs.py, fovs_decomposition.py, statistical helpers); the scripts/ directory contains one script per processing step. A top-level orchestrator, scripts/make_paper2.py, runs the complete pipeline from raw sections to the final figures and tables with idempotent per-output skipping. The full test suite passes (148 unit and integration tests); the ruff linter (PyFlakes-fatal rules, excluding unused local variables) passes on every commit. End-to-end wall-clock time from the raw sections is approximately 4–6 hours on a single 2024-era workstation, dominated by the SODA 5-day snapshot downloads; re-running from the archived reduced data takes under five minutes.

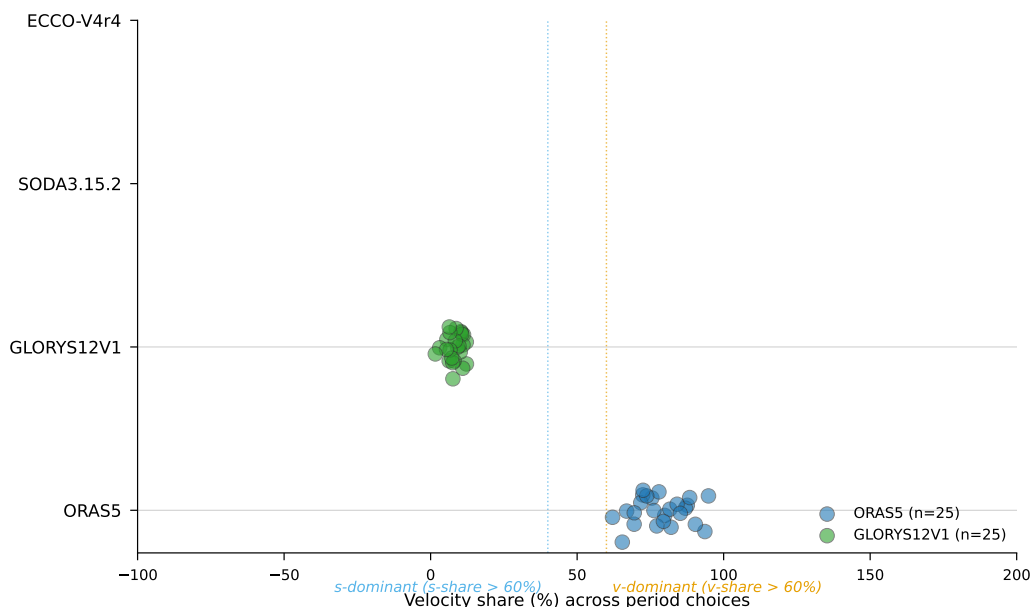


Figure S2: **Sensitivity of the mechanism classification to the choice of decomposition windows.** Velocity-share (f_v) distributions across 25 early/late period pairs spanning $t_{1,\text{end}} \in \{2001, 2003, 2005, 2007, 2009\}$ and $t_{2,\text{start}} \in \{2011, 2013, 2015, 2017, 2019\}$, always with $t_{1,\text{start}} = 1993$ and $t_{2,\text{end}}$ at each product's data end. The ORAS5 and GLORYS12V1 distributions never overlap; the mechanism disagreement between the two products survives every reasonable window choice and is not an artefact of the pre-registered 1993–2005 versus 2013–2025 pair used in the main text.

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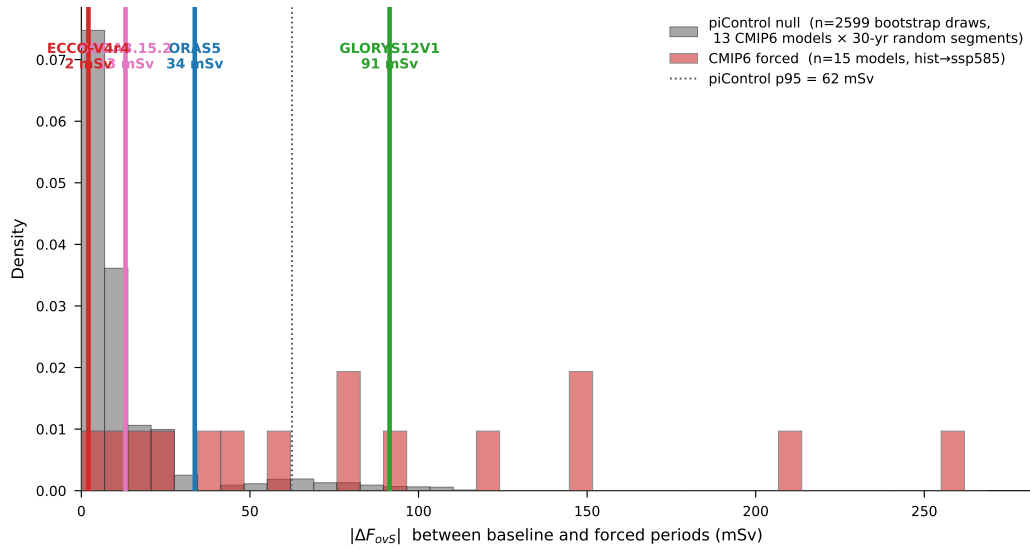


Figure S4: **Signal-to-noise test against piControl internal variability.** Pooled distribution of $|\Delta F_{OVS}|$ from 2600 random pairs of non-overlapping 30-year windows drawn from the piControl runs of 13 CMIP6 models (200 pairs per model). The 95th percentile is 22 mSv (vertical line). The forced reanalysis values for ORAS5 (31 mSv) and GLORYS12V1 (87 mSv) both exceed the 95th percentile, demonstrating that their mechanism attributions are signal-dominated rather than internal-variability artefacts. SODA 3.15.2 (14 mSv) sits between the median and the 95th percentile, consistent with the mixed classification it receives.

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